CHAPTER 4

Making Decisions

In the last chapter, we explored variables and how to declare them, their data types, assigning values, and performing simple operations using them. In this chapter, we will put the variables to work with some fundamental structures that the Arduino uses to make decisions and perform repetitive operations. This could include reading a switch or sensor and doing something based on its input, performing one or a set of operations for a defined number of times or until a different condition is met, or performing a range of operations depending on a range of conditions.

Our next project, Tilt Blink, adds a tilt switch to our RGB LED to form the basis for the rest of the chapter's in-depth discussions of the various statements for making decisions, such as the `if`, `for`, and `switch` statements and other control structures. We've also changed our code some, just to make things more interesting.

What's needed for this chapter:

- Arduino Uno
- 5mm RGB LED with common cathode
- Rolling ball tilt switch or other mechanical switch
- 1x 330 ohm , 2x 220 ohm, and 1x 10 kilohm ¼ watt resistor or similar
- Hookup wires
- Solderless breadboard

Project 3: Tilt Blink

For this project, we are building on our 7-Color Blink project from the last chapter with the addition of a simple tilt sensor. Let's say we want to quickly blink through the seven basic colors when our device is tilted or shaken, but otherwise turn off the LEDs when it's left alone. The tilt switch we are using to make this happen is a small metal tube with two wires coming out the bottom. Inside is a small metal ball that, when flat makes contact with the two wires closing the switch, but when tilted, the ball rolls away—opening the switch. This will be our way to turn on and off the RGB LED. Alternatively, any number of mechanical switches could be used instead if you can't find a tilt switch.